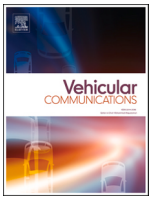
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Spectral- and energy-efficient power control for LEO satellite-assisted vehicular uplink spectrum sharing

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ABSTRACT

This paper investigates adaptive uplink power control for a low Earth orbit (LEO) satellite-assisted vehicular communication network, where a mobile satellite ground terminal reuses the same spectrum as nearby terrestrial base stations (BSs). Such satellite–terrestrial spectrum sharing can extend service coverage for vehicular and remote mobile users, but it may also produce harmful interference to terrestrial receivers. A spectral-efficiency (SE) maximization problem is first formulated under average transmit-power and average interference constraints. By applying the Lagrangian method, a closed-form interference-aware water-filling policy is derived, which adapts to both the desired LEO satellite channel and the strongest terrestrial interference channel. The framework is then extended to direct energy-efficiency (EE) maximization. The resulting fractional program is solved using Dinkelbach's method, and the EE-optimal power-control policy is derived in closed form for each Dinkelbach iteration. In addition, a hybrid learning-and-optimization benchmark is developed, where an AI-assisted power predictor is combined with one-step Dinkelbach refinement to reduce online computational complexity while approaching the EE-optimal benchmark. Simulation results reveal the transition between interference-limited and power-limited regimes. They also show that the SE-optimal policy achieves higher throughput in the high-threshold region, whereas the EE-optimal policy significantly improves EE by avoiding excessive transmit power. These results demonstrate the importance of objective-aware power control for balancing throughput, energy consumption, and terrestrial interference protection in LEO satellite-assisted vehicular uplink communications.

1. Introduction

1.1. Background and motivation

Vehicular communication networks are evolving toward ubiquitous, reliable, and energy-efficient connectivity for connected vehicles, intelligent transportation systems, logistics, public safety, and remote-area mobility services. Although terrestrial cellular networks can provide high-capacity coverage in urban and suburban areas, their service availability may degrade on rural roads, maritime routes, mountainous regions, and disaster-affected areas where terrestrial infrastructure is sparse or unavailable. Low Earth orbit (LEO) satellites provide a promising complementary infrastructure for extending vehicular and mobile connectivity beyond the coverage of terrestrial networks. Due to their relatively low orbital altitude, LEO satellites can offer lower propagation latency and improved link budgets compared with geostationary satellites, making them attractive for future integrated satellite–terrestrial vehicular communication systems [1–3].

Non-terrestrial networks (NTNs) have been actively studied in both standardization and research communities. In 3GPP, NTNs refer to networks or network segments that use space-borne or airborne platforms, including LEO, medium Earth orbit, geostationary orbit satellites, high-altitude platform systems, and unmanned aerial systems [4]. For vehicular and mobile users, LEO satellite access can improve service continuity when terrestrial coverage is intermittent. However, integrated satellite–terrestrial operation also introduces new interference-management challenges, especially when satellite and terrestrial systems reuse the same spectrum [5–7].

Spectrum sharing can improve spectral utilization, but it also creates inter-system interference coupling. In the uplink scenario considered in this paper, a mobile satellite ground terminal (GT), which may represent a vehicular or transportable user terminal, transmits data to a LEO satellite while nearby terrestrial BSs operate in the same frequency band. The uplink signal can therefore become harmful interference to terrestrial receivers. Since a vehicular or mobile terminal may move close to different terrestrial BSs over time, the strongest GT-to-BS interference channel becomes critical for terrestrial protection. This motivates adaptive power-control policies that account for both the desired LEO satellite link and the strongest terrestrial interference link.

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Cognitive satellite–terrestrial networking provides a useful framework for such coexistence. In a cognitive satellite–terrestrial network, one system may reuse the spectrum of another system as long as the imposed interference is controlled below a prescribed threshold [8–10]. For LEO satellite-assisted vehicular uplink communications, the power-control policy must not only improve the uplink throughput of the mobile terminal but also protect terrestrial BSs and avoid unnecessary energy consumption. Therefore, spectral efficiency and energy efficiency should be jointly investigated under the same interference-aware spectrum-sharing framework. It should be noted that the present work targets throughput-oriented uplink services in LEO-assisted vehicular communications, where sufficiently long codewords make the Shannon-capacity approximation appropriate. For ultra-reliable low-latency communication (URLLC) services employing short packets, finite-blocklength transmission models become more suitable and lead to different optimization structures. Such scenarios are discussed later as a potential extension of the proposed framework.

1.2. Related works

Resource allocation in integrated satellite–terrestrial and vehicular communication networks has been widely studied from different perspectives, including power allocation, beam scheduling, user association, interference coordination, energy efficiency, and learning-based optimization. Early works on cognitive satellite–terrestrial networks considered power allocation under quality-of-service and interference constraints [8]. Power control for real-time cognitive satellite–terrestrial applications was investigated in [9], where terrestrial protection constraints were explicitly considered. Optimal power control for cognitive LEO constellations sharing spectrum with terrestrial networks was studied in [10], showing that transmit-power adaptation is essential for satisfying interference requirements in satellite–terrestrial coexistence.

Energy efficiency has also become an important design objective in mobile, vehicular, satellite, and integrated networks. In practical vehicular or transportable terminals, energy consumption remains important because the communication unit may operate under a limited power supply or constrained on-board energy budget. For this reason, maximizing throughput alone may not be sufficient. Global energy-efficiency optimization for Ka-band multi-beam LEO satellite systems was investigated in [11]. Secure and energy-efficient transmission for rate-splitting multiple access (RSMA)-based cognitive satellite–terrestrial networks was studied in [12]. Energy-efficient optimization for RSMA-based integrated satellite–terrestrial networks under shadowed-Rician fading was considered in [13], which further highlights the importance of jointly considering multiple-access design, satellite fading, and energy-aware resource allocation. Energy-efficient power control for cognitive multi-beam satellite–terrestrial networks with randomly distributed terrestrial users was considered in [14], while energy-efficient power control under outdated CSI was studied in [15]. These works demonstrate the importance of energy-aware transmission in spectrum-sharing satellite and mobile communication systems.

Satellite channel modeling is another important aspect of performance analysis. Compared with terrestrial wireless channels, satellite channels may experience line-of-sight propagation, shadowing, atmospheric effects, and long-distance path loss. The shadowed-Rician model has been widely used to characterize land mobile satellite channels with both line-of-sight and scattered components [16]. Recent stochastic-geometry-based analyses have further investigated the impact of shadowed-Rician fading on satellite communication performance [17]. In cognitive satellite–terrestrial and vehicular NTN scenarios, accurate modeling of both the desired satellite channel and the terrestrial interference channels is necessary for reliable power-control design.

Fractional programming provides an important mathematical tool for EE maximization. Since EE is commonly defined as the ratio between the achievable rate and the consumed power, the resulting optimization problem naturally has a fractional structure. Dinkelbach's method is a classical approach for solving nonlinear fractional programs [18]. In wireless communications, fractional programming has been widely applied to energy-efficient power control, beamforming, and resource allocation problems [19,20]. This motivates the use of Dinkelbach-based optimization for the direct EE maximization problem considered in this paper.

Recently, artificial intelligence (AI) and machine learning have been investigated for satellite, vehicular, and NTN optimization. AI-based methods can learn complex mappings between channel states, mobility-related network states, and resource-allocation decisions, thereby reducing online computational burden in dynamic environments [21,22]. In RSMA-based integrated satellite–terrestrial networks, constrained deep reinforcement learning has been applied to EE optimization, where the original constrained resource-allocation problem is handled through a learning-based policy-gradient framework [23]. However, pure learning-based methods may suffer from approximation errors and constraint violations, especially when strict terrestrial interference protection is required. This motivates hybrid learning-and-optimization methods, where a learning model provides fast initialization and an optimization refinement step improves feasibility and performance.

1.3. Research gap

Although the above works provide important insights into satellite–terrestrial spectrum sharing and energy-efficient resource allocation, several gaps remain for LEO satellite-assisted vehicular uplink communications. First, many existing power-control studies mainly focus on spectral-efficiency or throughput maximization under interference constraints. While throughput maximization is important for high-data-rate vehicular services, it may allocate excessive transmit power when the interference threshold is loose. Due to the logarithmic rate function, the additional throughput gain obtained from high transmit power may become marginal, whereas the energy cost continues to increase. Therefore, the throughput-optimal operating point is not necessarily energy-efficient.

Second, direct EE maximization has a different mathematical structure from SE maximization. The EE objective is fractional and introduces an additional energy price into the power-allocation decision. Therefore, the optimal transmit power must account for the desired LEO satellite channel, the terrestrial interference channel, the average power budget, the interference threshold, and the energy cost of transmission. A unified comparison between SE-optimal and EE-optimal policies under the same LEO satellite-assisted vehicular uplink spectrum-sharing model is still limited.

Third, mobile or vehicular GTs may experience different terrestrial interference conditions as they move across the coverage area. When multiple terrestrial BSs coexist with the satellite uplink, the most vulnerable terrestrial receiver is often associated with the maximum GT-to-BS interference channel. Modeling the equivalent interference channel as the strongest terrestrial link can capture this protection requirement more accurately than using only an average interference channel.

Finally, optimization-based EE power control may involve iterative Dinkelbach updates and nested bisection for Lagrange multipliers. Although this provides a principled solution, it may be computationally demanding for online implementation in dynamic vehicular environments. Pure AI-based prediction can reduce online complexity but may introduce an optimality gap. This motivates a hybrid AI-assisted and optimization-refined approach for near-optimal low-complexity EE power control.

Another practical challenge in LEO-assisted vehicular communications is that multiple mobile terminals may simultaneously access the same satellite beam while sharing spectrum with terrestrial networks. In such cases, transmit-power allocation becomes coupled with user scheduling, beam assignment, and inter-user interference management. Before addressing these network-level interactions, it is essential to understand the optimal power-control behavior of a single uplink, which

serves as the fundamental building block for more sophisticated multi-terminal resource-allocation frameworks.

1.4. Main contributions

The main contributions of this paper are summarized as follows:

- A LEO satellite-assisted vehicular uplink spectrum-sharing framework is formulated under average transmit-power and average interference constraints.
- The equivalent terrestrial interference channel is modeled as the strongest GT-to-BS channel, thereby capturing the dominant terrestrial protection requirement in dense BS deployments.
- A closed-form SE-optimal power-control policy is derived, and its interference-aware water-filling structure is characterized.
- A direct EE maximization problem is formulated and transformed into a tractable auxiliary problem using Dinkelbach's method.
- The EE-optimal power-control policy is derived in closed form for each Dinkelbach iteration, revealing the role of an additional energy price in the effective water level.
- A hybrid AI-assisted EE power-control benchmark with one-step Dinkelbach refinement is developed, showing that a lightweight learning model can provide useful initialization while one refinement step restores near-optimal EE performance.
- The SE-optimal, EE-optimal, and AI-assisted refined policies are compared in terms of throughput, EE, average transmit power, and average interference, highlighting the tradeoff among spectral efficiency, energy efficiency, terrestrial protection, and online implementation complexity.
- Although the analysis focuses on a single GT-satellite uplink, the proposed framework establishes a tractable analytical baseline that can be integrated into future multi-terminal scheduling and resource-allocation algorithms.

1.5. Organization

The rest of this paper is organized as follows. Section 2 presents the system and channel models. Section 3 introduces the performance metrics and optimization problems. Section 4 derives the SE-optimal power-control policy. Section 5 develops the EE-optimal power-control policy using Dinkelbach's method and presents the AI-assisted one-step refinement benchmark. Section 6 provides numerical results, including the SE–EE tradeoff, three-dimensional EE behavior, and AI-assisted refinement performance. Finally, Section 7 concludes the paper.

Notation: Throughout this paper, X denotes the desired satellite-link channel power gain, Y_m denotes the GT-to-BS interference-channel power gain, and $Y = \max_{1\le m\le M} Y_m$ denotes the equivalent strongest interference channel. The transmit power is denoted by $q(X, Y)$, while $P_{\max}$ and $I_{\max}$ denote the average transmit-power budget and average interference threshold, respectively. The notation $[z]^+ = \max(z, 0)$ is used throughout.

2. System and channel models

We consider a LEO satellite-assisted vehicular uplink spectrum-sharing network, as illustrated in Fig. 1. The network consists of one mobile satellite ground terminal (GT), one LEO satellite, and M surrounding terrestrial base stations (BSs). The GT may represent a vehicular, maritime, or transportable user terminal that relies on LEO satellite connectivity when terrestrial coverage is unavailable or insufficient. To establish the satellite uplink, the GT employs a directional antenna with beam-tracking capability to communicate with the serving LEO satellite while limiting undesired radiation toward terrestrial systems. The same licensed frequency band is simultaneously reused by the terrestrial cellular network, enabling spectrum sharing between the satellite and terrestrial systems. Consequently, although the desired satellite transmission benefits from the directional beam, the GT may still generate interference to nearby terrestrial BSs through sidelobes and propagation leakage. Therefore, adaptive uplink transmit-power control is required to maximize the communication performance while satisfying the average transmit-power and terrestrial interference constraints.

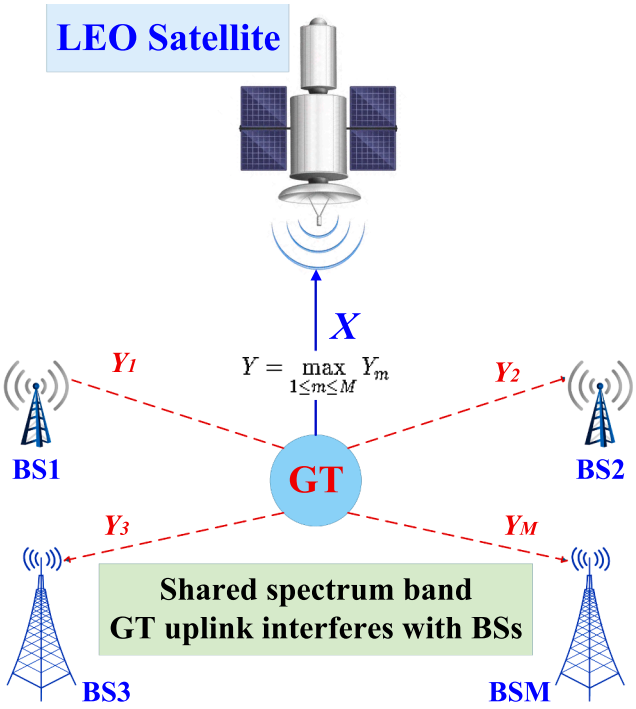


Fig. 1. Physical system model of the considered LEO satellite-assisted vehicular uplink spectrum-sharing network. A mobile ground terminal (GT) communicates with the serving LEO satellite through the desired satellite-link channel gain X while causing interference to M surrounding terrestrial base stations (BSs) through the interference channel gains $\{Y_m\}_{m=1}^{M}$. To guarantee terrestrial-network protection, the equivalent interference channel is modeled as the strongest interference link, $Y = \max_{1\le m\le M} Y_m$.

Throughout this paper, frequency-division duplexing (FDD) is assumed for both the satellite and terrestrial systems. The proposed power-control framework operates at the GT based on the estimated satellite-link and terrestrial interference channel gains without requiring modifications to the existing terrestrial network infrastructure.

Let X denote the instantaneous channel power gain of the desired satellite uplink from the GT to the LEO satellite. Let Y_m denote the instantaneous interference-channel power gain from the GT to the mth terrestrial BS, where $m \in \{1, 2, \dots, M\}$. Since the terrestrial BS experiencing the strongest interference dominates the protection requirement, the equivalent terrestrial interference channel is modeled as

$$Y = \max_{1\le m\le M} Y_m. \tag{1}$$

Let q be the GT transmit power and let σ^2 be the receiver noise power at the LEO satellite. The received SNR is

$$\gamma = \frac{qX}{\sigma^2}. \tag{2}$$

The instantaneous spectral efficiency is

$$R(q) = \log_2\left(1 + \frac{qX}{\sigma^2}\right). \tag{3}$$

The Shannon-capacity expression is adopted to facilitate analytical derivations. This assumption is appropriate for long-codeword transmission, whereas finite-blocklength formulations may be considered for latency-critical vehicular services [24,25].

The desired satellite channel gain is modeled as

$$X = \Omega_X |h_s|^2, \tag{4}$$

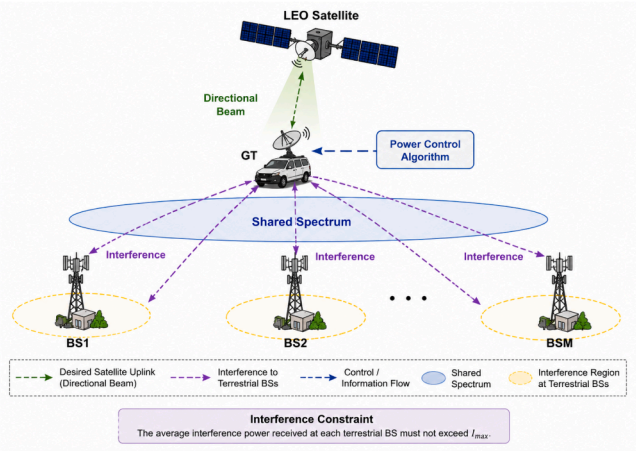


Fig. 2. Overall system architecture of the proposed LEO satellite-assisted vehicular uplink spectrum-sharing system. The mobile ground terminal (GT) estimates the satellite and terrestrial channel gains and executes the proposed power-control algorithm locally to determine the uplink transmit power while protecting nearby terrestrial base stations (BSs).

where Ω_X captures the average satellite-link scaling factor and h_s is the small-scale fading coefficient. The interference channel from the GT to the mth terrestrial BS is modeled as

$$Y_m = \Omega_Y |h_m|^2, \tag{5}$$

where Ω_Y is the average terrestrial interference-channel scaling factor. Under Rayleigh fading, $|h_m|^2$ follows an exponential distribution. As M increases, the maximum channel gain Y statistically increases, requiring more conservative transmit-power allocation to satisfy the terrestrial protection constraint.

The GT must satisfy the average transmit-power constraint

$$\mathbb{E}[q(X,Y)] \le P_{\max}, \tag{6}$$

where $P_{\max}$ is the maximum allowable average transmit power. The average interference caused to the terrestrial system must also satisfy

$$\mathbb{E}[q(X,Y)Y] \le I_{\max}, \tag{7}$$

where $I_{\max}$ is the maximum allowable average interference power.

2.1. System architecture

While Fig. 1 illustrates the physical topology of the considered spectrum-sharing network, Fig. 2 shows the functional architecture adopted for practical implementation. The proposed power-control algorithm is executed locally at the mobile ground terminal (GT), which periodically acquires the estimated satellite-link channel gain and the strongest terrestrial interference channel gain. These channel estimates are then used to determine the uplink transmit power before each transmission interval.

The computed transmit power is applied to the directional satellite uplink to maximize the communication performance while satisfying the interference constraint imposed on the terrestrial network. Since the optimization is performed entirely at the GT, the proposed framework does not require any modification to the existing terrestrial infrastructure or additional coordination among terrestrial BSs.

Throughout this paper, frequency-division duplexing (FDD) is assumed for both the satellite and terrestrial systems. The proposed architecture relies only on locally available channel estimates and can therefore be integrated into practical LEO satellite-assisted vehicular communication systems with limited signaling overhead.

2.2. Practical CSI acquisition and discussion on finite-blocklength transmission

The proposed power-control policies require the GT to obtain estimates of both the desired satellite-link channel gain and the strongest terrestrial interference channel gain. Throughout this paper, the analytical derivations assume sufficiently accurate short-term channel state information (CSI), which serves as an upper-bound benchmark commonly adopted in optimization-based resource-allocation studies.

In practical LEO satellite-assisted vehicular systems, the desired satellite-link CSI can be acquired through pilot-assisted channel estimation together with satellite ephemeris information and beam-tracking procedures. Since the orbital trajectory of LEO satellites is deterministic, the large-scale channel variations can be accurately predicted, while the remaining small-scale fading is estimated using uplink pilots.

For terrestrial protection, the GT does not require the instantaneous CSI of every surrounding BS. Instead, the strongest GT-to-BS interference channel can be estimated through network-assisted feedback, radio-environment maps, spectrum sensing, or location-assisted interference prediction. Consequently, the effective interference channel is represented by the estimated strongest channel gain.

Accordingly, the proposed power-control policies can be directly implemented using the estimated channel gains $(\hat{X}, \hat{Y})$ instead of the ideal values (X, Y). Although CSI estimation errors and feedback delays may slightly reduce optimality, the proposed analytical framework remains applicable as a benchmark and provides useful design insights for practical implementations. Therefore, the analytical derivations presented in the remainder of this paper employ the ideal channel gains (X, Y) for notational simplicity, whereas the practical implementation uses their corresponding estimates $(\hat{X}, \hat{Y})$. This distinction does not alter the optimization framework or the derived closed-form power-control policies.

Furthermore, the proposed framework adopts the Shannon-capacity expression to enable tractable analytical optimization under long-codeword transmission. For latency-sensitive vehicular applications, finite-blocklength (FBL) transmission provides a more accurate characterization of the achievable rate [24,25], which can be approximated as

$$R_{\mathrm{FBL}} = \log_2(1+\gamma) - \sqrt{\frac{V(\gamma)}{n}} Q^{-1}(\varepsilon), \tag{8}$$

where $V(\gamma)$ denotes the channel dispersion, n is the coding blocklength, and ε is the decoding error probability. Incorporating the above FBL expression into the proposed optimization framework would require reformulating both the spectral-efficiency and energy-efficiency optimization problems. Such an extension is beyond the scope of the present work and will be investigated in future research.

Although the analysis is presented at the link level, the proposed architecture is compatible with practical NTN deployments where beam management, mobility management, and satellite handover are performed by higher network layers.

3. Performance metrics and optimization problems

For a given power-control policy $q(X,Y)$, the average spectral efficiency is

$$\bar{R} = \mathbb{E}\left[\log_2\left(1 + \frac{q(X,Y)X}{\sigma^2}\right)\right]. \tag{9}$$

The average transmit power and average interference are given by

$$\bar{P}_{\mathrm{tx}} = \mathbb{E}[q(X,Y)], \tag{10}$$

$$\bar{I} = \mathbb{E}[q(X,Y)Y]. \tag{11}$$

The EE is defined as

$$\eta_{\mathrm{EE}} = \frac{\bar{R}}{\bar{P}_{\mathrm{tx}} + P_c}, \tag{12}$$

where P_c is the circuit power consumption.

Remark 1. The SE and EE definitions adopted in this paper are based on Shannon capacity, which is appropriate for sufficiently long packets. For finite-blocklength transmission, the achievable rate can be replaced by the normal approximation without changing the overall optimization framework, although the resulting power-control solution may no longer admit the same closed-form structure.

3.1. SE maximization problem

The SE-oriented problem maximizes the average spectral efficiency under average transmit-power and average interference constraints:

$$\max_{q(X,Y)\geq 0} \quad \mathbb{E}\left[\log_2\left(1+\frac{q(X,Y)X}{\sigma^2}\right)\right] \tag{13a}$$

$$\text{s.t.} \quad \mathbb{E}[q(X,Y)] \leq P_{\max}, \tag{13b}$$

$$\mathbb{E}[q(X,Y)Y] \leq I_{\max}. \tag{13c}$$

This is a convex optimization problem in the sense that a concave objective is maximized over an affine feasible set.

3.2. EE maximization problem

The EE-oriented problem directly maximizes the EE:

$$\max_{q(X,Y)\geq 0} \quad \frac{\mathbb{E}\left[\log_2\left(1+\frac{q(X,Y)X}{\sigma^2}\right)\right]}{\mathbb{E}[q(X,Y)]+P_c} \tag{14a}$$

$$\text{s.t.} \quad \mathbb{E}[q(X,Y)] \leq P_{\max}, \tag{14b}$$

$$\mathbb{E}[q(X,Y)Y] \leq I_{\max}. \tag{14c}$$

Unlike (13), problem (14) has a fractional objective and is solved through fractional programming.

3.3. Applicability of the Shannon capacity model

The optimization problems in (13) and (14) are formulated based on the Shannon-capacity expression

$$R = \log_2(1+\gamma), \tag{15}$$

which is appropriate for throughput-oriented LEO satellite-assisted vehicular communications employing sufficiently long channel codewords. Under this assumption, the achievable rate depends solely on the received signal-to-noise ratio (SNR), leading to tractable spectral-efficiency (SE) and energy-efficiency (EE) optimization problems with closed-form power-control policies.

For latency-sensitive vehicular services employing short packets, the achievable rate is more accurately characterized by the finite-blocklength (FBL) model introduced in Section 2.2. Incorporating the FBL rate expression would modify the objective functions of both the SE- and EE-oriented optimization problems, and the resulting power-control policies would generally no longer admit the same closed-form solutions derived in this paper. Therefore, the proposed framework should be regarded as an analytical benchmark for long-codeword transmission, whereas extending the optimization framework to finite-blocklength communications remains an important direction for future research.

4. SE-oriented power control

To solve (13), two non-negative Lagrange multipliers are introduced: β for the average transmit-power constraint and α for the average interference constraint. For a given channel realization (X,Y), the Lagrangian term related to q is

$$\mathcal{L}(q) = \log_2\left(1+\frac{qX}{\sigma^2}\right) - \beta q - \alpha q Y. \tag{16}$$

Taking the derivative with respect to q and applying the KKT condition yields

$$\frac{1}{\ln 2}\frac{X}{\sigma^2+qX} = \beta + \alpha Y. \tag{17}$$

Proposition 1. *The optimal solution of the SE-oriented problem in (13) is given by*

$$q^{\star}_{\mathrm{SE}}(X,Y) = \left[\frac{1}{\ln(2)(\beta+\alpha Y)} - \frac{\sigma^2}{X}\right]^+, \tag{18}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are selected to satisfy the active average interference and transmit-power constraints.

Proof. See Appendix A. □

Eq. (18) has an interference-aware water-filling structure. A strong desired channel X reduces the inverse channel-quality penalty σ^2/X, whereas a strong interference channel Y increases the effective water-level penalty $\beta+\alpha Y$. If the interference constraint is inactive, then $\alpha = 0$, and the policy reduces to conventional power-limited water filling. If the interference constraint is active, then $\alpha > 0$, and the policy explicitly reduces the transmit power in channel states that would cause strong terrestrial interference. In practical implementations, the GT performs power adaptation using the estimated channel gains $\hat{X}$ and $\hat{Y}$ obtained through channel estimation rather than the ideal channel gains X and Y. Accordingly, the online implementation follows

$$q_{\mathrm{SE}}(\hat{X},\hat{Y}) = \left[\frac{1}{\ln(2)(\beta+\alpha\hat{Y})} - \frac{\sigma^2}{\hat{X}}\right]^+, \tag{19}$$

where $\hat{X}$ and $\hat{Y}$ denote the estimated desired satellite-link and strongest terrestrial interference channel gains, respectively. The analytical derivation of Eq. (18) therefore serves as an upper-bound benchmark under accurate short-term CSI, while the same closed-form structure can be directly applied using practical CSI estimates (Algorithm 1).

Algorithm 1 Bisection-based SE-oriented power control.

```
Generate Monte Carlo samples of X and Y.
Set α = 0 and find β by bisection from E[q] ≤ P_max.
Compute q using (18).
if E[qY] ≤ I_max then
    Use the power-only solution and stop.
else
    Initialize lower and upper bounds for α.
    while the outer bisection has not converged do
        Set α to the midpoint of the current bounds.
        For this α, find β by inner bisection.
        Compute q using (18).
        if E[qY] > I_max then
            Increase α.
        else
            Decrease α.
        end if
    end while
end if
Compute R̄ = E[log2(1 + qX/σ^2)].
```

5. EE-oriented power control

5.1. Dinkelbach-based EE-optimal power control

The fractional problem in (14) is solved using Dinkelbach's method. For a given non-negative parameter ν, define the auxiliary problem

$$F(\nu) = \max_{q(X,Y)\geq 0} E\left[\log_2\left(1+\frac{q(X,Y)X}{\sigma^2}\right)\right] - \nu\big(E[q(X,Y)]+P_c\big) \tag{20a}$$

$$\text{s.t.} \quad \mathbb{E}[q(X,Y)] \le P_{\max}, \tag{20b}$$

$$\mathbb{E}[q(X,Y)Y] \le I_{\max}. \tag{20c}$$

The optimal EE $\nu^\star$ is achieved when $F(\nu^\star)=0$. At convergence,

$$\nu^\star = \frac{\bar{R}^\star}{\bar{P}_{\text{tx}}^\star + P_c} = \eta_{\text{EE}}^\star. \tag{21}$$

For a fixed ν, two non-negative Lagrange multipliers are introduced: λ for the average transmit-power constraint and μ for the average interference constraint. For each channel realization (X,Y), the Lagrangian term related to q is

$$\mathcal{L}_{\text{EE}}(q) = \log_2\left(1+\frac{qX}{\sigma^2}\right) - \nu q - \lambda q - \mu q Y. \tag{22}$$

Taking the derivative and applying the KKT condition gives the EE-optimal power-control policy for each Dinkelbach iteration.

Proposition 2. *For a fixed Dinkelbach parameter ν, the optimal solution of the auxiliary EE problem in (20) is*

$$q_{\text{EE}}^\star(X,Y) = \left[\frac{1}{\ln(2)(\nu+\lambda+\mu Y)} - \frac{\sigma^2}{X}\right]^+, \tag{23}$$

where $\lambda \ge 0$ and $\mu \ge 0$ are selected to satisfy the active average transmit-power and interference constraints.

Proof. See Appendix B. □

Compared with the SE-optimal policy, the EE-optimal policy contains the additional term ν, which acts as an energy price. A larger ν penalizes power consumption more strongly and therefore reduces the allocated transmit power. For practical implementation, the GT computes the transmit power from the estimated channel gains $(\hat{X},\hat{Y})$, yielding

$$q_{\text{EE}}(\hat{X},\hat{Y}) = \left[\frac{1}{\ln(2)(\nu+\lambda+\mu\hat{Y})} - \frac{\sigma^2}{\hat{X}}\right]^+. \tag{24}$$

Hence, the proposed EE-oriented policy preserves exactly the same computational structure as Eq. (23), with only the ideal channel gains replaced by their practical estimates. This enables direct implementation using pilot-assisted CSI estimation without modifying the underlying optimization framework.

Algorithm 2 Dinkelbach-based EE-oriented power control.

1: Generate Monte Carlo samples of X and Y.
2: Initialize $\nu^{(0)}=0$ and tolerance $\epsilon>0$.
3: **repeat**
4: For a given $\nu^{(k)}$, solve (20).
5: Determine λ and μ by nested bisection.
6: Compute $q_{\text{EE}}^{(k)}(X,Y)$ using (23).
7: Compute $\bar{R}^{(k)} = \mathbb{E}[\log_2(1+q_{\text{EE}}^{(k)}X/\sigma^2)]$.
8: Compute $\bar{P}^{(k)} = \mathbb{E}[q_{\text{EE}}^{(k)}]$.
9: Update $\nu^{(k+1)} = \bar{R}^{(k)}/(\bar{P}^{(k)}+P_c)$.
10: **until** $\left|\bar{R}^{(k)} - \nu^{(k)}(\bar{P}^{(k)}+P_c)\right| \le \epsilon$
11: Output $q_{\text{EE}}^\star(X,Y)$ and $\eta_{\text{EE}}^\star = \nu^{(k+1)}$.

5.2. AI-assisted EE power control: network architecture, training procedure, and one-step refinement

Although the Dinkelbach-based algorithm provides an optimization-based solution to the EE maximization problem, it requires multiple outer Dinkelbach iterations together with nested bisection searches for the Lagrange multipliers. Although this approach provides the optimal EE solution, it may introduce non-negligible computational latency in dynamic LEO-assisted vehicular communications. To reduce the online computational burden while preserving near-optimal EE performance, a hybrid AI-assisted power-control framework is developed (Fig. 3).

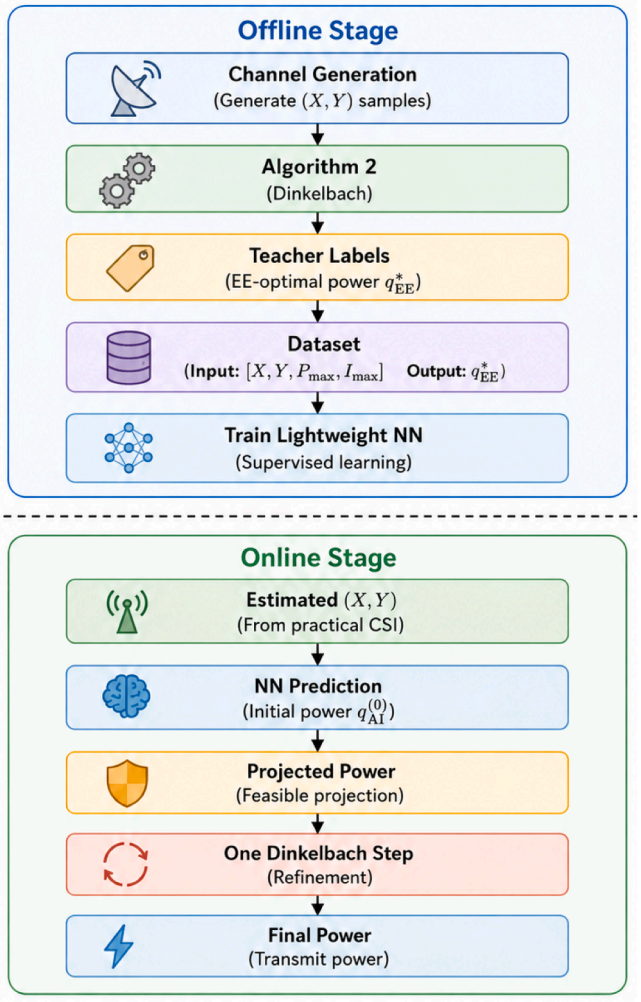


Fig. 3. Workflow of the proposed AI-assisted EE power-control framework. During the offline stage, channel samples are generated and Algorithm 2 produces the EE-optimal transmit powers, which are used as teacher labels for neural-network training. During the online stage, the estimated channel gains $(\hat{X},\hat{Y})$ are processed by the trained network to predict an initial transmit power, which is then projected onto the feasible region and refined through one Dinkelbach iteration to obtain the final transmit power.

The proposed AI-assisted framework employs the Dinkelbach-based EE-optimal policy as the teacher model. A lightweight fully connected neural network is adopted owing to its low inference complexity and suitability for latency-sensitive vehicular applications. The network consists of one input layer, two hidden layers with ReLU activation functions, and one linear output layer.

The input feature vector is

$$\mathbf{s} = [X, Y, P_{\max}, I_{\max}]^T, \tag{25}$$

where X and Y denote the desired satellite-link and strongest terrestrial interference channel gains, respectively. The network predicts the initial transmit power

$$q_{\text{AI}}^{(0)}(X,Y). \tag{26}$$

All input features are normalized into the interval $[0,1]$ before training to improve numerical stability.

The training dataset is generated offline using Algorithm 2. For each randomly generated channel realization and system configuration, the Dinkelbach algorithm computes the corresponding EE-optimal transmit power. Therefore, each training sample is represented by

$$(X, Y, P_{\max}, I_{\max}) \to q_{\text{EE}}^\star(X,Y). \tag{27}$$

Approximately 10^5 channel realizations are generated. The dataset is randomly divided into training, validation, and testing subsets using an 80%/10%/10% split.

The network parameters are optimized by minimizing the mean-squared error (MSE)

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left(q_i^{\text{pred}} - q_i^{\text{opt}}\right)^2, \tag{28}$$

where q_i^{pred} and q_i^{opt} denote the predicted and EE-optimal transmit powers for the ith training sample, respectively. The network is trained using the Adam optimizer with an initial learning rate of 10^{-3}. Training continues until the validation loss converges.

During online operation, only one forward inference is required to predict the initial transmit power $q_{\text{AI}}^{(0)}(X,Y)$. Since the predicted solution may contain approximation errors, it is first projected onto the feasible average transmit-power and average interference region. The resulting policy is then used to compute the initial Dinkelbach parameter

$$\nu_{\text{AI}}^{(0)} = \frac{\mathbb{E}\left[\log_2\left(1+\frac{q_{\text{AI}}^{(0)}X}{\sigma^2}\right)\right]}{\mathbb{E}[q_{\text{AI}}^{(0)}]+P_c}. \tag{29}$$

Using $\nu_{\text{AI}}^{(0)}$, only one auxiliary Dinkelbach refinement step is performed. The refined transmit power is

$$q_{\text{AI-ref}}(X,Y) = \left[\frac{1}{\ln(2)\left(\nu_{\text{AI}}^{(0)}+\lambda+\mu Y\right)} - \frac{\sigma^2}{X}\right]^+. \tag{30}$$

Here, λ and μ are selected such that the average transmit-power and average interference constraints are satisfied. Consequently, the neural network provides a fast initialization, whereas the subsequent refinement restores near-optimal EE performance while maintaining strict constraint satisfaction. Since the neural-network prediction already provides an initial transmit power that is close to the EE-optimal solution, the one-step refinement typically achieves a near-converged solution after only a single auxiliary Dinkelbach update. This substantially reduces the online computational burden compared with the conventional Dinkelbach algorithm while preserving nearly the same EE performance (Algorithm 3).

Algorithm 3 AI-assisted EE power control with one-step Dinkelbach refinement.

Offline Stage:
Generate channel samples $(X, Y, P_{\max}, I_{\max})$.
Run Algorithm 2 to obtain the EE-optimal transmit powers $q_{\text{EE}}^{\star}(X,Y)$.
Generate teacher labels $\{(X, Y, P_{\max}, I_{\max}), q_{\text{EE}}^{\star}\}$.
Normalize the input features.
Train the lightweight neural network using the generated dataset.
Store the trained network parameters.

Online Stage:
Estimate the current channel gains $(\hat{X}, \hat{Y})$.
Normalize the estimated input features.
Perform one neural-network inference to predict $q_{\text{AI}}^{(0)}(\hat{X},\hat{Y})$.
Project the predicted transmit power onto the feasible average transmit-power and interference region.
Compute the initial Dinkelbach parameter $\nu_{\text{AI}}^{(0)}$.
Perform one Dinkelbach refinement step.
Transmit using the refined power $q_{\text{AI-ref}}(\hat{X},\hat{Y})$.

Compared with Algorithm 2, the proposed AI-assisted framework eliminates repeated Dinkelbach iterations during online operation. The online computational complexity is dominated by a single forward inference of the neural network,

$$\mathcal{O}\left(\sum_{l} n_l n_{l+1}\right), \tag{31}$$

where n_l denotes the number of neurons in the lth layer. By contrast, the Dinkelbach-based method requires multiple outer iterations together with nested bisection searches. Therefore, the proposed AI-assisted framework considerably reduces the online computational effort. Since only one forward inference is required during online operation, the proposed AI-assisted framework is well suited for latency-sensitive vehicular communications that require rapid transmit-power adaptation under dynamically varying channel conditions. The performance evaluation presented in Section 6 further demonstrates that the AI-assisted framework achieves EE performance close to the optimization-based benchmark.

Table 1
Comparison of online implementation complexity of the considered power-control algorithms.

Method	Required Information	Online Procedure
SE-optimal	Channel samples/statistics	Outer dual-variable update with bisection search
EE-optimal (Dinkelbach)	Channel samples/statistics	Outer Dinkelbach iterations with nested bisection searches
AI-assisted	Offline teacher dataset	Single neural-network forward inference
AI-assisted + one-step refinement	Offline teacher dataset	Neural-network inference followed by one Dinkelbach refinement step

Table 2
Training configuration of the proposed AI-assisted neural network.

Parameter	Value
Input features	X, Y, $P_{\max}$, $I_{\max}$
Output	$q_{\text{AI}}^{(0)}$
Hidden layers	2
Activation function	ReLU
Output activation	Linear
Loss function	Mean squared error (MSE)
Optimizer	Adam
Initial learning rate	10^{-3}
Dataset size	10^5 samples
Training / Validation / Testing	80% / 10% / 10%

Table 1 summarizes the online implementation procedures of the considered power-control methods. The SE- and EE-oriented optimization algorithms require iterative numerical optimization during online operation, whereas the EE-oriented scheme additionally performs Dinkelbach iterations. In contrast, the AI-assisted policy replaces the iterative optimization procedure with a single neural-network forward inference. The AI-assisted policy with one-step refinement further performs one auxiliary Dinkelbach update after the neural-network prediction to improve the transmit-power allocation.

6. Simulation results and discussion

In this section, numerical simulations are presented to evaluate the SE-optimal, EE-optimal, and AI-assisted refined policies. Unless otherwise specified, the average transmit-power budgets are $P_{\max} = 25$ dBm, 20 dBm, and 15 dBm. The noise power is set to −94 dBm, corresponding to $\sigma^2 = 10^{(-94-30)/10}$ W. The considered setup corresponds to a mobile or vehicular GT operating in a shared satellite–terrestrial spectrum environment, where the strongest GT-to-BS interference link represents the most vulnerable terrestrial receiver during mobile uplink transmission.

The SE-optimal policy is compared with a reference scheme in which the instantaneous strongest interference channel Y is approximated by

Table 3
Simulation parameters.

Parameter	Value
Vehicle scenario	Urban vehicular uplink
Spectrum sharing mode	Underlay spectrum sharing
Frequency reuse	Full frequency reuse
Duplex mode	FDD
Noise power, σ^2	−94 dBm
Maximum transmit power, $P_{\max}$	15, 20, 25 dBm
Average interference threshold, $I_{\max}$	−65 to 5 dBm
Number of terrestrial BSs, M	1–10 (or fixed as specified)
Circuit power, P_c	20 dBm
Satellite channel model	Shadowed-Rician / scaled fading
Terrestrial interference channel	Rayleigh fading
Monte Carlo samples	10^4–2×10^5

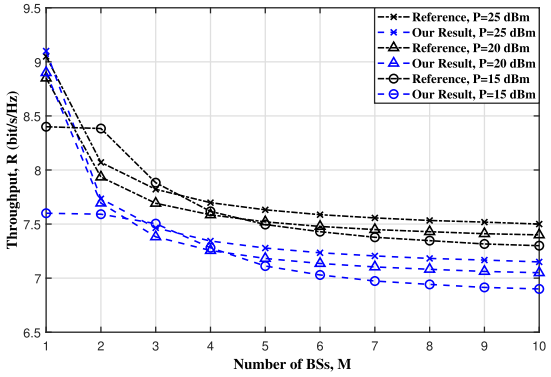


Fig. 4. Average throughput versus the number of surrounding terrestrial BSs.

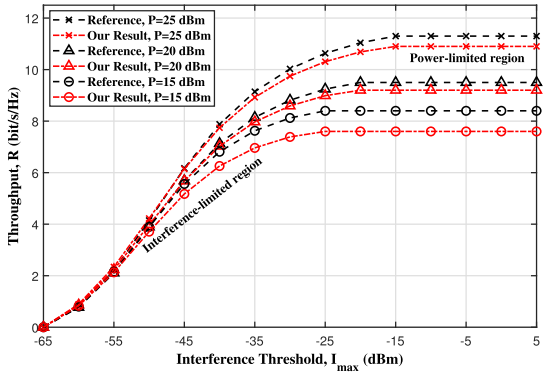


Fig. 5. Average throughput versus the interference threshold.

its average value $\mathbb{E}[Y]$ when computing the transmit power. The obtained power allocation is then evaluated under the actual channel realizations. This benchmark represents an approximation-based policy that does not fully exploit instantaneous interference-channel information. The training configuration of the proposed AI-assisted neural network and the main simulation parameters are summarized in (Tables 2 and 3), respectively. The EE-optimal policy is then compared with the SE-optimal policy to quantify the throughput–energy tradeoff.

6.1. SE-optimal throughput performance

Fig. 4 shows the average throughput as a function of the number of surrounding terrestrial BSs. As M increases, the equivalent interference channel $Y = \max_m Y_m$ becomes stronger. Consequently, the GT must reduce its transmit power to satisfy the average interference constraint, resulting in a decreasing throughput trend. The throughput decreases rapidly when the first few BSs are added and then decreases more slowly

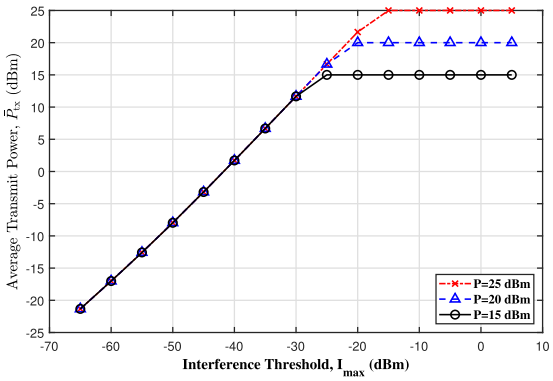


Fig. 6. Average transmit power of the SE-optimal policy versus interference threshold.

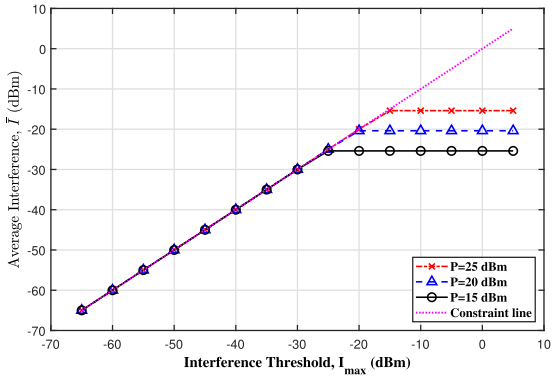


Fig. 7. Average interference of the SE-optimal policy versus interference threshold.

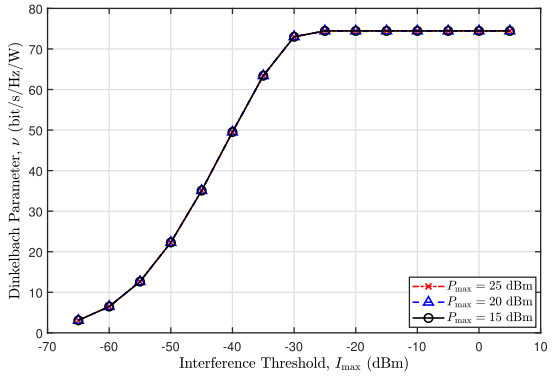


Fig. 8. Optimal Dinkelbach parameter $\nu^\star$ versus interference threshold. Since $\nu^\star = \eta_{\mathrm{EE}}^\star$ at convergence, the curve represents the maximum achievable EE.

as M further increases because the incremental increase of the maximum of multiple fading channels becomes less significant for large M.

Fig. 5 shows the SE-optimal average throughput versus the interference threshold $I_{\max}$. When $I_{\max}$ is small, the GT is strongly restricted by terrestrial protection, and the throughput is close to zero. As $I_{\max}$ increases, the feasible transmit-power region becomes larger, and the throughput increases. When $I_{\max}$ becomes sufficiently large, the interference constraint becomes inactive and the system enters the power-limited region, where increasing $I_{\max}$ no longer provides significant throughput improvement.

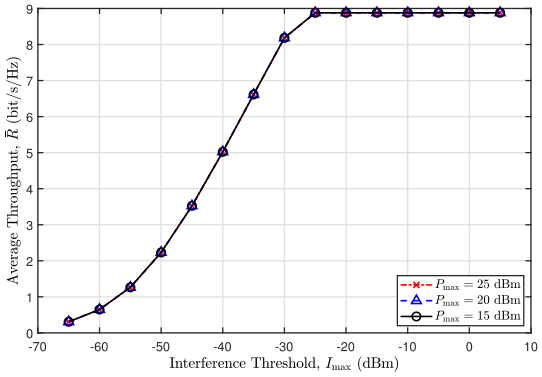


Fig. 9. Average throughput of the EE-optimal policy versus interference threshold.

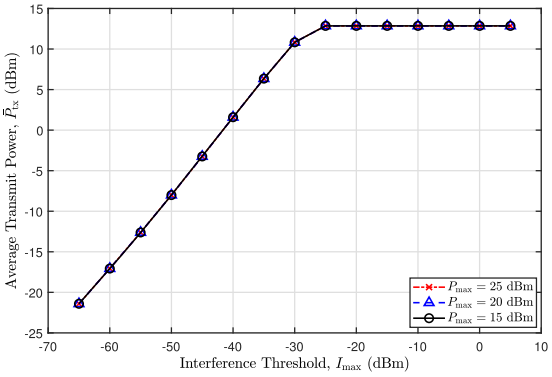


Fig. 10. Average transmit power of the EE-optimal policy versus interference threshold.

6.2. SE-optimal power and interference behavior

Fig. 6 shows the average transmit power of the SE-optimal policy. In the interference-limited region, the GT must keep its transmit power low to protect terrestrial BSs. As $I_{\max}$ increases from −65 dBm to approximately −30 dBm, the average transmit power increases from about −21 dBm to around 12 dBm. When $I_{\max}$ is further increased, the curves saturate at their corresponding power budgets, confirming the transition to the power-limited region.

Fig. 7 shows the average interference $\bar{I} = \mathbb{E}[qY]$ of the SE-optimal policy. In the low-threshold region, the achieved average interference closely follows the constraint line, indicating that the interference constraint is active. When $I_{\max}$ is further increased, the curves deviate from the constraint line and saturate because the transmit-power constraint becomes dominant.

6.3. EE-optimal performance

Fig. 8 shows the optimal Dinkelbach parameter $\nu^\star$ versus the interference threshold $I_{\max}$. Since $\nu^\star = \eta_{\text{EE}}^\star$ at convergence, this curve directly represents the maximum achievable EE of the EE-optimal policy. The EE increases rapidly as $I_{\max}$ is relaxed from −65 dBm to approximately −30 dBm and then saturates around 74–75 bit/s/Hz/W. The curves for different $P_{\max}$ values almost overlap, indicating that the EE-optimal transmit power is below all considered power budgets in the high-threshold region.

Fig. 9 shows the average throughput achieved by the EE-optimal policy. The throughput increases from approximately 0.3 bit/s/Hz at $I_{\max} = -65$ dBm to about 8.2 bit/s/Hz at $I_{\max} = -30$ dBm, and then saturates around 8.9 bit/s/Hz. Unlike the SE-optimal policy, the EE-optimal policy stops increasing transmit power once the EE-optimal operating point is reached.

Fig. 10 shows the average transmit power of the EE-optimal policy. The transmit power increases in the interference-limited region and saturates around 13 dBm when the EE-optimal operating point is reached. This value is lower than all considered power budgets, showing that the EE-optimal policy does not necessarily use the full available power.

The consistent trends observed in Figs. 8–10 can also be explained directly from the structure of the EE-optimal solution. At Dinkelbach convergence, substituting $\nu = \nu^\star$ into (23) yields

$$q_{\text{EE}}^\star(X,Y) = \left[\frac{1}{\ln(2)(\nu^\star + \lambda + \mu Y)} - \frac{\sigma^2}{X}\right]^+, \tag{32}$$

where λ and μ are the Lagrange multipliers associated with the average transmit-power and interference constraints, respectively. When $I_{\max}$ is small, the interference constraint is active and $\mu > 0$. Relaxing $I_{\max}$ reduces the corresponding interference price μ, increases the effective water level, and therefore permits larger transmit powers over favorable channel states. Consequently, both the average throughput and the optimal EE, represented by $\nu^\star = \eta_{\text{EE}}^\star$, increase with $I_{\max}$. At the same time, the average transmit power also increases because progressively more channel states become energetically worthwhile for transmission.

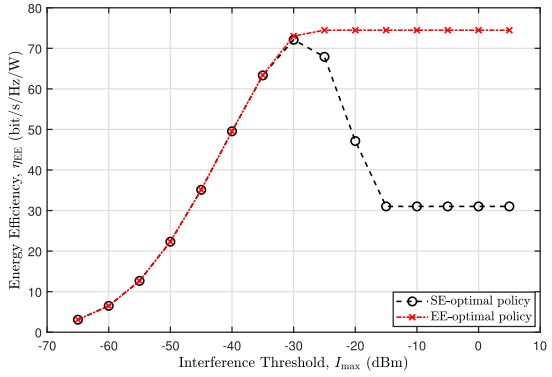


Fig. 11. EE comparison between the SE-optimal and EE-optimal policies.

Once $I_{\max}$ becomes sufficiently large, however, the interference constraint becomes inactive and $\mu \to 0$. The solution then approaches the unconstrained EE-optimal operating point, at which any further increase in transmit power would provide a marginal throughput gain smaller than its associated energy cost. Hence, $\nu^\star$, the average throughput, and the average transmit power all converge to saturation values. Moreover, if this unconstrained EE-optimal average transmit power is below $P_{\max}$, the transmit-power constraint is also inactive, i.e., $\lambda = 0$. This explains why the curves corresponding to different $P_{\max}$ values nearly coincide in Figs. 8–10: the EE optimum is determined primarily by the intrinsic rate–power tradeoff rather than by the available maximum power budget.

This behavior is also consistent with the complementary-slackness conditions. In the interference-limited regime, $\mu > 0$ and

$$\mathbb{E}\left[q_{\text{EE}}^\star(X,Y)Y\right] = I_{\max}, \tag{33}$$

whereas after the interference constraint becomes non-binding, $\mu = 0$. Similarly, $\lambda = 0$ whenever

$$\mathbb{E}\left[q_{\text{EE}}^\star(X,Y)\right] < P_{\max}. \tag{34}$$

6.4. Comparison between SE-optimal and EE-optimal policies

Fig. 11 compares the EE achieved by the SE-optimal and EE-optimal policies. In the low-threshold region, both policies achieve almost the same EE because both are strongly interference-limited. However, when $I_{\max}$ becomes sufficiently large, the EE-optimal policy saturates around

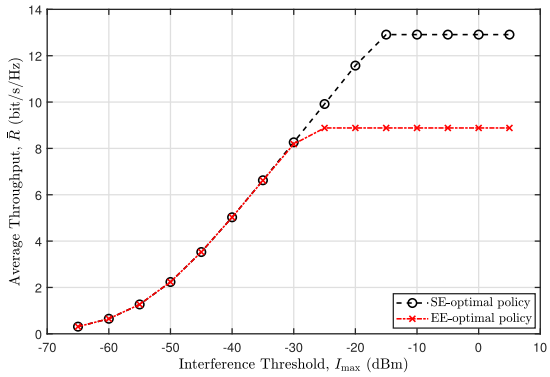


Fig. 12. Average throughput comparison between the SE-optimal and EE-optimal policies.

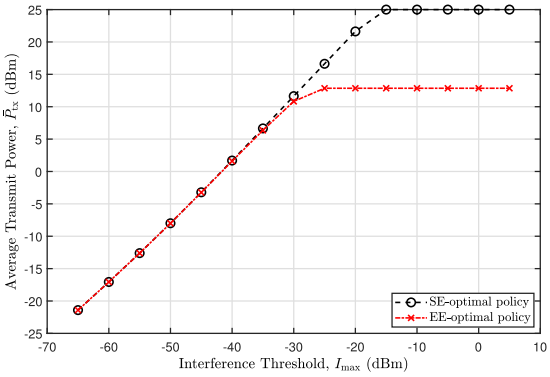


Fig. 13. Average transmit-power comparison between the SE-optimal and EE-optimal policies.

74–75 bit/s/Hz/W and maintains this value, whereas the SE-optimal policy decreases and eventually saturates around 31 bit/s/Hz/W. The decrease of the SE-optimal policy occurs because it continues to increase transmit power for marginal throughput gain.

Fig. 12 compares the average throughput of the two policies. In the low-threshold region, both policies are almost identical. In the high-threshold region, the SE-optimal policy continues to increase transmit power and reaches a higher saturation throughput of approximately 13 bit/s/Hz, whereas the EE-optimal policy saturates earlier at approximately 8.9 bit/s/Hz. This confirms the throughput–energy tradeoff.

Fig. 13 compares the average transmit power of the SE-optimal and EE-optimal policies. When $I_{\max}$ is small, both policies use almost the same transmit power because the interference constraint dominates. When $I_{\max}$ is large, the SE-optimal policy saturates at the power budget, around 25 dBm in this case, whereas the EE-optimal policy saturates around 13 dBm. This explains why the EE-optimal policy achieves higher EE in the high-threshold region.

Fig. 14 shows the three-dimensional EE surface of the EE-optimal policy as a function of the interference threshold $I_{\max}$ and the transmit-power budget $P_{\max}$. In the interference-limited region, $\eta_{\text{EE}}^{\star}$ increases rapidly as $I_{\max}$ is relaxed because the feasible transmit-power region becomes larger. When $I_{\max}$ becomes sufficiently large, the surface becomes nearly flat, indicating that the EE-optimal operating point has been reached. In this region, increasing $P_{\max}$ provides little additional gain because the EE-optimal policy does not necessarily use the full available power budget.

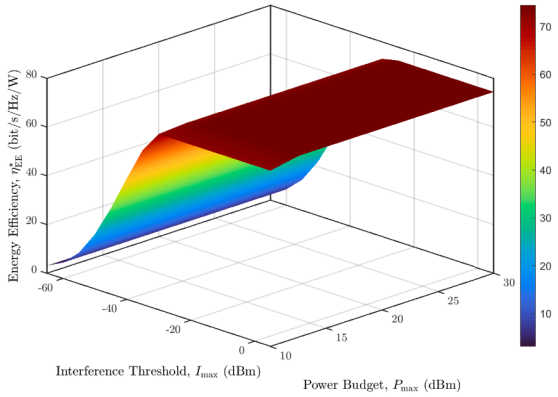


Fig. 14. Three-dimensional EE surface of the EE-optimal policy. The plateau in the high-threshold region indicates that increasing $P_{\max}$ no longer improves the maximum EE.

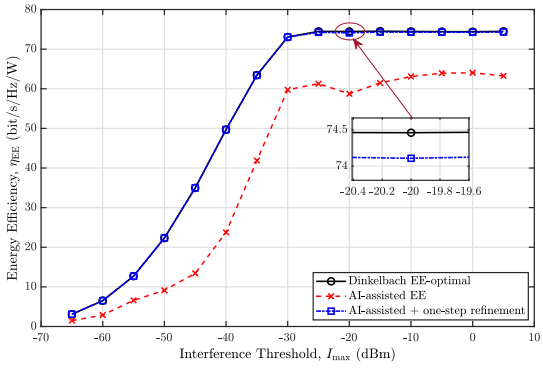


Fig. 15. EE comparison among the Dinkelbach-based EE-optimal policy, the AI-assisted policy, and the AI-assisted policy with one-step Dinkelbach refinement.

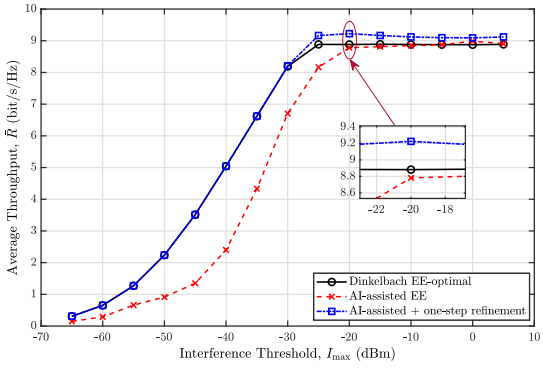


Fig. 16. Average throughput comparison among the Dinkelbach-based EE-optimal policy, the AI-assisted policy, and the AI-assisted policy with one-step Dinkelbach refinement.

6.5. AI-assisted EE power-control benchmark

To evaluate the potential of low-complexity online implementation, an AI-assisted EE power-control benchmark is considered. The Dinkelbach-based EE-optimal policy is used as a teacher model to generate offline training labels. In the simulations, an extreme learning machine is used as a lightweight learning model because it provides closed-form training and low online inference complexity. Since the pure AI-assisted policy may suffer from an approximation gap, a one-step Dinkelbach refinement is further applied. Specifically, the EE value obtained from the AI-assisted policy is used as an initial Dinkelbach

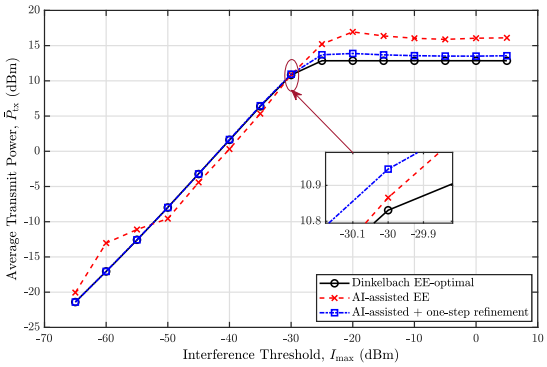


Fig. 17. Average transmit-power comparison among the Dinkelbach-based EE-optimal policy, the AI-assisted policy, and the AI-assisted policy with one-step Dinkelbach refinement.

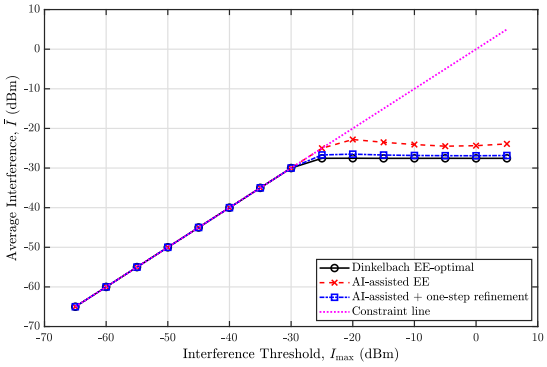


Fig. 18. Average interference comparison among the Dinkelbach-based EE-optimal policy, the AI-assisted policy, and the AI-assisted policy with one-step Dinkelbach refinement.

parameter, and one auxiliary EE maximization step is performed to refine the transmit-power allocation.

Fig. 15 compares the EE of the Dinkelbach-based EE-optimal policy, the pure AI-assisted policy, and the AI-assisted policy with one-step Dinkelbach refinement. The pure AI-assisted policy captures the increasing trend of the EE-optimal solution, but it exhibits a visible optimality gap, especially in the high-threshold region. In contrast, the one-step refined policy almost overlaps with the Dinkelbach-based EE-optimal curve over the entire range of $I_{\max}$. This shows that a lightweight AI predictor can provide a useful initialization, while the one-step refinement effectively restores near-optimal EE performance.

Fig. 16 shows the corresponding average throughput. The pure AI-assisted policy achieves lower throughput in the interference-limited and transition regions, which explains its lower EE. After one-step refinement, the achieved throughput becomes very close to that of the Dinkelbach-based EE-optimal policy. In the high-threshold region, the refined policy may achieve a slightly higher throughput due to a small difference in the refined transmit-power level. Nevertheless, the corresponding EE remains nearly identical to the Dinkelbach-based benchmark, indicating that the refinement step reaches an operating point close to the EE optimum.

Fig. 17 compares the average transmit power. The pure AI-assisted policy tends to use higher power in the high-threshold region, which reduces its EE due to the marginal rate gain of the logarithmic throughput function. The one-step refined policy corrects this behavior and brings the transmit power close to the EE-optimal level. This confirms that the refinement step is effective in recovering the energy-efficient operating point.

Fig. 18 shows the average interference. In the low-threshold region, all schemes follow the interference constraint because terrestrial protection dominates the feasible power allocation. When $I_{\max}$ becomes large, the refined policy remains below the constraint line and closely follows the Dinkelbach-based EE-optimal interference level. Therefore, the proposed AI-assisted refinement preserves the interference protection requirement while achieving near-optimal EE performance.

Fig. 19 evaluates the learning capability of the proposed AI-assisted EE power-control framework. Fig. 19(a) presents the training and validation mean squared error (MSE) versus the training epoch. Both curves decrease rapidly during the early training stage and gradually converge after approximately 40 epochs to an MSE of about 10^{-4}. No evident overfitting can be observed because the training and validation curves closely match throughout the training process, indicating good generalization capability of the proposed neural network.

Fig. 19(b) compares the transmit power predicted by the proposed neural network with the EE-optimal teacher labels generated by Algorithm 2. Most prediction samples closely follow the ideal 45° reference line, achieving a coefficient of determination of $R^2 = 0.9955$ and a root mean squared error (RMSE) of 3.17×10^{-4} W. These results indicate that the proposed lightweight neural network provides an accurate approximation of the optimization-based power-allocation policy while requiring only one forward inference during online operation.

Fig. 19(c) presents the cumulative distribution function (CDF) of the absolute prediction error. More than 90% of the prediction errors are below approximately 10^{-4} W, whereas the maximum prediction error remains on the order of 10^{-3} W. These results demonstrate that the approximation error introduced by the proposed neural network is small relative to the considered transmit power range. Consequently, the AI-assisted predictor provides a reliable initialization for the subsequent one-step Dinkelbach refinement, allowing the proposed framework to achieve EE performance close to the optimization-based benchmark with substantially lower online computational effort.

Fig. 20 illustrates the impact of imperfect CSI on the normalized energy efficiency (EE), where the CSI quality is characterized by the correlation coefficient ρ. As expected, the EE of all considered schemes gradually decreases as ρ decreases from 1 to 0.6 because channel-estimation errors and feedback delays reduce the accuracy of the transmit-power allocation. Specifically, the normalized EE of the SE-oriented policy decreases from 1 to approximately 0.923, corresponding to a degradation of about 7.7%. In comparison, the proposed EE-oriented policy exhibits a smaller performance loss of approximately 4.3%, confirming that jointly optimizing spectral efficiency and power consumption improves robustness against CSI mismatch. Furthermore, the proposed AI-assisted policy with one-step refinement consistently achieves the highest normalized EE over the entire CSI range. Even under severe CSI degradation ($\rho = 0.6$), the normalized EE remains approximately 0.967, corresponding to only about 3.3% degradation relative to the perfect-CSI case. These results demonstrate that the proposed AI-assisted framework effectively preserves the performance of the EE-optimal solution while maintaining strong robustness under practical CSI uncertainty. These observations indicate that the proposed power-control framework is not overly sensitive to CSI imperfections, suggesting that the analytical policies derived under ideal CSI remain effective benchmarks for practical LEO satellite-assisted vehicular systems with imperfect channel estimation.

The presented numerical results correspond to the representative single-terminal uplink considered throughout this paper. In practical satellite-assisted vehicular networks, multiple terminals may compete for the same satellite resources. Nevertheless, the proposed power-control policy can be employed as a per-user optimization module after user scheduling, thereby preserving its practical applicability in larger-scale systems.

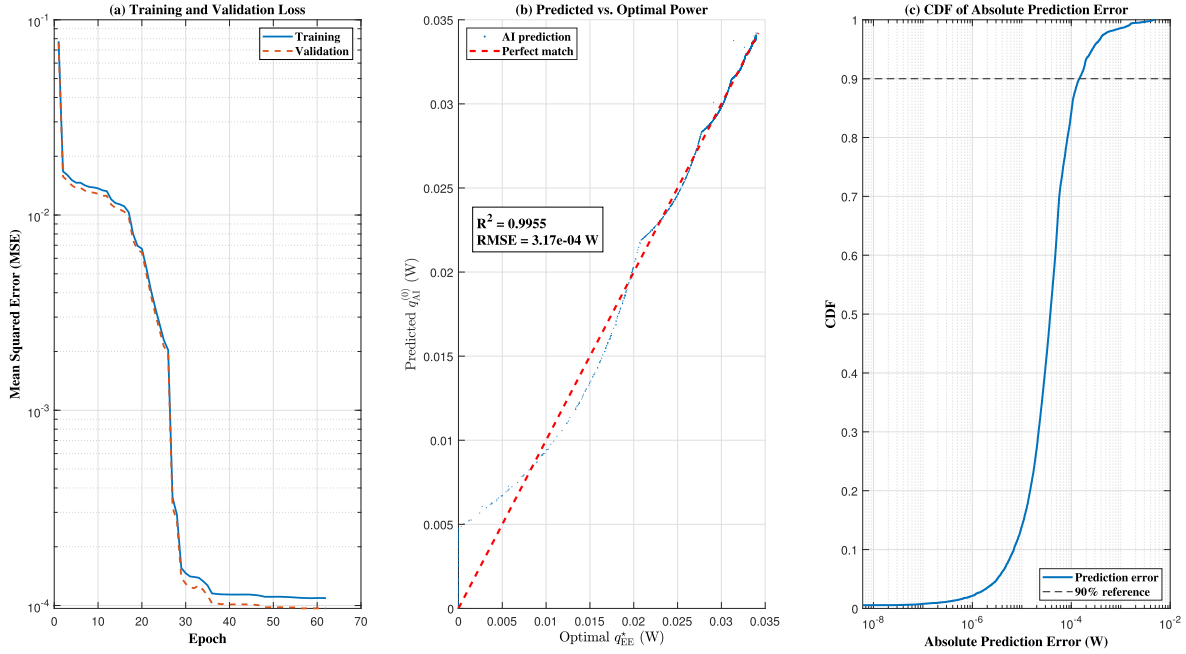


Fig. 19. Performance evaluation of the proposed AI-assisted EE power-control framework. (a) Training and validation mean squared error (MSE) versus training epoch. (b) Comparison between the transmit power predicted by the proposed neural network and the EE-optimal teacher labels generated by Algorithm 2, achieving $R^2 = 0.9955$ and an RMSE of 3.17×10^{-4} W. (c) Cumulative distribution function (CDF) of the absolute prediction error.

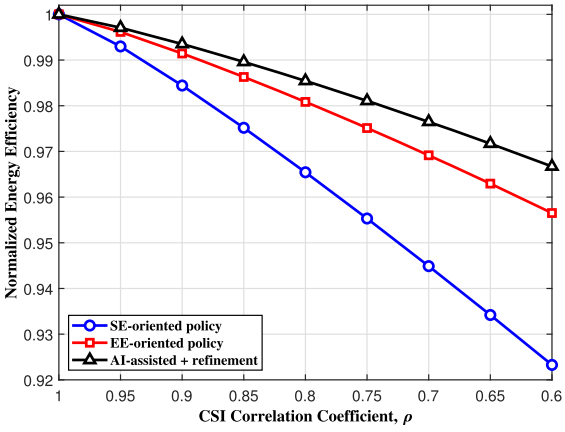


Fig. 20. Impact of imperfect CSI on the normalized energy efficiency of different power-control policies. The CSI quality is characterized by the correlation coefficient ρ, where $\rho = 1$ corresponds to perfect CSI. As ρ decreases, all schemes experience EE degradation due to less accurate channel estimates. The proposed EE-oriented policy and the AI-assisted one-step refinement scheme remain more robust than the SE-oriented policy over the entire CSI range.

7. Conclusion

This paper investigated adaptive power control for a LEO satellite-assisted vehicular uplink spectrum-sharing network. An SE-oriented power-control problem was first formulated under average transmit-power and average interference constraints. By applying the Lagrangian method, a closed-form interference-aware water-filling policy was derived, which adapts to both the desired LEO satellite channel and the strongest terrestrial interference channel. A direct EE maximization problem was then formulated and solved using Dinkelbach's method. The resulting EE-optimal policy introduces an additional energy price into the effective water level, thereby avoiding excessive power consumption when the marginal throughput gain becomes small.

Simulation results demonstrated the transition between the interference-limited and power-limited regimes. In the high-threshold region, the SE-optimal policy achieves higher throughput by exploiting the available power budget, whereas the EE-optimal policy maintains significantly higher EE by selecting a more energy-conscious transmit-power level. Furthermore, the AI-assisted policy with one-step Dinkelbach refinement closely approaches the Dinkelbach-based EE-optimal benchmark while preserving the average interference constraint.

The proposed lightweight neural network accurately learns the mapping between the channel-state information and the EE-optimal power-allocation policy, enabling near-optimal online performance with only a single forward inference. The numerical results also demonstrated that the proposed AI-assisted framework exhibits good generalization capability without noticeable overfitting and remains more robust than the conventional SE-oriented policy under imperfect CSI.

The derived closed-form policies can be directly implemented using estimated channel gains obtained through practical pilot-assisted CSI acquisition, making the proposed framework suitable for practical LEO satellite-assisted vehicular communications with latency-sensitive power adaptation.

Future work will investigate robust power-control designs under imperfect and delayed CSI, finite-blocklength communications, and more realistic high-mobility LEO vehicular scenarios that explicitly account for channel estimation errors, feedback latency, and rapidly time-varying channels. These results demonstrate that the proposed framework provides a practical balance between analytical tractability, computational efficiency, and implementation feasibility for future LEO satellite-assisted vehicular networks.

CRediT authorship contribution statement

Huu Q. Tran: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Appendix A. Proof of the SE-optimal power-control policy

For a given channel realization (X, Y), the Lagrangian term of the SE-oriented problem is

$$\mathcal{L}(q) = \log_2\left(1 + \frac{qX}{\sigma^2}\right) - \beta q - \alpha q Y. \tag{35}$$

The first derivative is

$$\frac{\partial \mathcal{L}}{\partial q} = \frac{1}{\ln 2}\frac{X}{\sigma^2 + qX} - \beta - \alpha Y. \tag{36}$$

The second derivative is

$$\frac{\partial^2 \mathcal{L}}{\partial q^2} = -\frac{1}{\ln 2}\frac{X^2}{(\sigma^2 + qX)^2} < 0, \tag{37}$$

for $X > 0$. Hence, the Lagrangian is strictly concave in q. Setting the first derivative to zero gives

$$\frac{1}{\ln 2}\frac{X}{\sigma^2 + qX} = \beta + \alpha Y. \tag{38}$$

Solving for q and projecting onto the non-negative domain yields (18).

Appendix B. Proof of the EE-optimal power-control policy

For a fixed Dinkelbach parameter ν, the channel-wise Lagrangian of the EE-oriented auxiliary problem is

$$\mathcal{L}_{\mathrm{EE}}(q) = \log_2\left(1 + \frac{qX}{\sigma^2}\right) - \nu q - \lambda q - \mu q Y. \tag{39}$$

The first derivative is

$$\frac{\partial \mathcal{L}_{\mathrm{EE}}}{\partial q} = \frac{1}{\ln 2}\frac{X}{\sigma^2 + qX} - \nu - \lambda - \mu Y. \tag{40}$$

Setting the derivative to zero gives

$$\frac{1}{\ln 2}\frac{X}{\sigma^2 + qX} = \nu + \lambda + \mu Y. \tag{41}$$

Solving for q and applying the non-negativity constraint gives (23). Since the logarithmic rate is concave and the constraints are affine, the KKT solution is globally optimal for each Dinkelbach iteration.

Appendix C. Distribution of the strongest interference channel

Assume that $Y_m = \Omega_Y |h_m|^2$, where $h_m \sim \mathcal{CN}(0,1)$. Then Y_m is exponentially distributed with mean Ω_Y, and its CDF is

$$F_{Y_m}(y) = 1 - \exp\left(-\frac{y}{\Omega_Y}\right), \quad y \geq 0. \tag{42}$$

Since $Y = \max_{1\leq m\leq M} Y_m$, the CDF of Y is

$$F_Y(y) = \Pr(Y \leq y) \tag{43}$$

$$= \prod_{m=1}^{M} F_{Y_m}(y) \tag{44}$$

$$= \left(1 - \exp\left(-\frac{y}{\Omega_Y}\right)\right)^M. \tag{45}$$

Differentiating $F_Y(y)$ gives

$$f_Y(y) = \frac{M}{\Omega_Y}\exp\left(-\frac{y}{\Omega_Y}\right)\left(1 - \exp\left(-\frac{y}{\Omega_Y}\right)\right)^{M-1}, \quad y \geq 0. \tag{46}$$

This distribution shifts toward larger values as M increases, explaining why dense terrestrial deployments require more conservative satellite uplink transmission.

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